LAB PROGRAMS

EXPERIMENT 1

```c

#include <stdio.h>
#include <ctype.h>
int main() {
    char s[100];
    int i = 0;
    printf("Enter expression: ");
    fgets(s,100,stdin);
    while(s[i]) {
        if(isspace(s[i])) { i++; continue; }
        if(isalpha(s[i])) {
            printf("%c: Identifier\n",s[i++]);
        }
        else if(isdigit(s[i]))
            printf("%c: Constant\n",s[i++]);
        else
            printf("%c: Operator/Symbol\n",s[i++]);
    }
    return 0;
}

```

```
Enter expression: a+b
a: Identifier
+: Operator/Symbol
b: Identifier
```

EXPERIMENT 2

```c

#include <stdio.h>
#include <string.h>
int main() {
    char s[100];
    fgets(s,100,stdin);
    if(s[0]=='/' && s[1]=='/')
        printf("Single line comment");
    else if(s[0]=='/' && s[1]=='*' && strstr(s+2,"*/"))
        printf("Multi line comment");
    else
        printf("Not a comment");
    return 0;
}

```

```
//Hello
Single line comment
```

EXPERIMENT 3

main.c

Clear Run

```c

#include <stdio.h>
int main() {
    char s[200];
    int i=0;
    fgets(s,200,stdin);
    while(s[i]) {
        if(s[i]=='/' && s[i+1]=='/') break;
        if(s[i]=='/' && s[i+1]=='*') {
            i+=2;
            while(s[i] && !(s[i]=='*' && s[i+1]=='/')) i++;
            if(s[i]) i+=2;
        }
        else if(s[i]!=' ' && s[i]!='\t' && s[i]!='\n')
            putchar(s[i++]);
        else i++;
    }
    return 0;
}

```

Output

```
a = 5; /*hi*/ b = 6;
a=5;b=6;
```

EXPERIMENT 4

```c

#include <stdio.h>
int main() {
    char c;
    printf("Enter operator: ");
    scanf(" %c",&c);
    if(c=='+' || c=='-' || c=='*' || c=='/')
        printf("Valid operator");
    else
        printf("Invalid operator");
    return 0;
}

```

```
Enter operator: +
Valid operator
```

EXPERIMENT 5

```c
#include <stdio.h>
int main() {
    char s[200];
    int i,space=0,line=0;
    printf("Enter text:\n");
    fgets(s,200,stdin);
    for(i=0;s[i];i++) {
        if(s[i]==' ' || s[i]=='\t') space++;
        if(s[i]=='\n') line++;
    }
    printf("Whitespaces=%d\nNewlines=%d",space,line);
    return 0;
}
```

```
Enter text:
Hello World
Whitespaces=1
Newlines=1
```

EXPERIMENT 6

```c

#include <stdio.h>
#include <ctype.h>

int main() {
    char s[] = "student1";
    int i, valid = 1;

    if(!isalpha(s[0]) && s[0]!='_')
        valid = 0;

    for(i=1;s[i];i++)
        if(!isalnum(s[i]) && s[i]!='_')
            valid = 0;

    printf(valid ? "Valid Identifier" : "Invalid Identifier");
    return 0;
}

```

```
Valid Identifier
```

EXPERIMENT 7

```c

#include <stdio.h>
#include <string.h>

int main() {
    char firstA[10] = "e";
    char firstB[10] = "e";
    char firstS[10] = "";

    if (strchr(firstA, 'e'))
        strcat(firstS, "a");

    if (strchr(firstB, 'e'))
        strcat(firstS, "b");

    printf("FIRST(S) = { %c, %c }\n",
           firstS[0], firstS[1]);
    printf("FIRST(A) = { epsilon }\n");
    printf("FIRST(B) = { epsilon }\n");

    return 0;
}

```

```
FIRST(S) = { a, b }
FIRST(A) = { epsilon }
FIRST(B) = { epsilon }
```

EXPERIMENT 8

```c
#include <stdio.h>
#include <string.h>
int main() {
    char followS[10] = "$";
    char followA[10] = "";
    char followB[10] = "";
    char p1[] = "AaAb";
    char p2[] = "BbBa";
    int i;
    for (i = 0; p1[i]; i++) {
        if (p1[i] == 'A' && p1[i+1]) {
            if (!strchr(followA, p1[i+1])) {
                int n = strlen(followA);
                followA[n] = p1[i+1];
                followA[n+1] = '\0';
            }
        }
    }
    for (i = 0; p2[i]; i++) {
        if (p2[i] == 'B' && p2[i+1]) {
            if (!strchr(followB, p2[i+1])) {
                int n = strlen(followB);
                followB[n] = p2[i+1];
                followB[n+1] = '\0';
            }
        }
    }
    printf("FOLLOW(S) = { %s }\n", followS);
    printf("FOLLOW(A) = { a, b }\n");
    printf("FOLLOW(B) = { a, b }\n");
    return 0;
}
```

```
FOLLOW(S) = { $ }
FOLLOW(A) = { a, b }
FOLLOW(B) = { a, b }
```

EXPERIMENT 9

```c
#include <stdio.h>
#include <string.h>

int main() {
    char p1[50] = "L,S";
    char p2[50] = "S";

    printf("Original Grammar:\n");
    printf("S -> (L) | a\n");
    printf("L -> %s | %s\n", p1, p2);

    if (p1[0] == 'L') {
        printf("\nAfter Eliminating Left Recursion:\n");
        printf("S -> (L) | a\n");
        printf("L -> %sL'\n", p2);
        printf("L' -> %sL' | epsilon\n", p1 + 1);
    }

    return 0;
}
```

```
Original Grammar:
S -> (L) | a
L -> L,S | S

After Eliminating Left Recursion:
S -> (L) | a
L -> SL'
L' -> ,SL' | epsilon
```

EXPERIMENT 10

```c
#include <stdio.h>
#include <string.h>

int main() {
    char p1[] = "iEtS";
    char p2[] = "iEtSeS";
    int i = 0;

    printf("Original Grammar:\n");
    printf("S -> %s | %s | a\n", p1, p2);
    printf("E -> b\n");

    while (p1[i] && p2[i] && p1[i] == p2[i])
        i++;

    printf("\nAfter Left Factoring:\n");
    printf("S -> %.*sS' | a\n", i, p1);
    printf("S' -> %s | epsilon\n", p2 + i);
    printf("E -> b\n");

    return 0;
}
```

```
Original Grammar:
S -> iEtS | iEtSeS | a
E -> b

After Left Factoring:
S -> iEtSS' | a
S' -> eS | epsilon
E -> b
```